Data Science Intern at Data Glacier

Project: Hate Speech Detection using Transformers (Deep Learning)

Week 8: Deliverable

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Table of Contents:

1. Project Plan

Weeks	Date	plan
Weeks 07	May 19, 2025	Problem Description, Business understanding, Data Intake Report
Weeks 08	May 26, 2025	Data Preprocessing
Weeks 09	June 2, 2025	Data Cleansing and Transformation
Weeks 10	June 9, 2025	EDA performed
Weeks 11	June 16, 2025	EDA Presentation and Proposed Modelling Technique
Weeks 12	June 23, 2025	Model Selection and Model Building
Weeks 13	June 30, 2025	Final Submission (Report + Code + Presentation)

2. Problem Description

Hate speech poses a significant challenge on digital platforms, often manifesting as verbal, written, or behavioral communication targeting individuals or groups based on religion, ethnicity, nationality, race, gender, or other identity traits. With the rise of social media usage, particularly platforms like Twitter, monitoring and mitigating such harmful content has become essential.

This project aims to build an automated hate speech detection system using advanced Natural Language Processing (NLP) techniques and transformer-based deep learning models. The task is framed as a binary text classification problem, where the model learns to distinguish between hateful and non-hateful tweets. The goal is to create a reliable and scalable model capable of supporting content moderation efforts by detecting hate speech in real time.

3. Business Understanding

Social media platforms are essential tools for global communication but also serve as mediums for the spread of hate speech and abusive content. The unchecked proliferation of such content can lead to social unrest, legal issues, and reputational damage for platform providers. Businesses operating these platforms are increasingly under pressure to ensure user safety and compliance with regulations on digital conduct.

By deploying an automated hate speech detection system, companies can significantly enhance their moderation capabilities. This not only helps maintain a safer online environment but also improves user trust and brand integrity. From a data science perspective, this project addresses key business goals such as minimizing manual moderation efforts, reducing harmful interactions, and supporting ethical AI practices in content governance.

4. Data Preprocessing

In this section, we describe the text preprocessing techniques applied to prepare the dataset for machine learning models. Raw text data typically contains various inconsistencies, noise, and irrelevant components that can negatively affect the performance of NLP-based models. Therefore, a sequence of cleaning steps was carried out to normalize and refine the input tweets.

4.1 Text Cleaning

Text data collected from social media platforms such as Twitter often contains unwanted elements like user mentions, hyperlinks, emojis, and non-standard formatting. Cleaning such data is essential for ensuring better feature extraction and model performance. The steps below summarize our cleaning pipeline:

4.1.1 Lowercasing

All characters in the text were converted to lowercase. This normalization step ensures that words like *"Hate"* and *"hate"* are treated identically. Without this transformation, these words would be considered different features in the model, leading to unnecessary sparsity and dimensionality in the feature space.

4.1.2 Removing User Mentions

Twitter handles or user mentions (e.g., *@username*) were removed. These references are unique to individual users and do not contribute semantically to the classification task. Their removal helps avoid model distraction from non-informative tokens.

4.1.3 Removing URLs

URLs (web links) were stripped from the text. Since the task focuses on analyzing the sentiment or intent of the written content, web addresses—which usually lead to external content—do not provide meaningful context and are treated as noise.

4.1.4 Removing Special Characters and Digits

Special characters (e.g., `@`, `#`, `!`, `%`) and numeric digits were eliminated. These characters typically do not carry linguistic meaning in the context of sentiment or hate speech detection. A regular expression-based filter was applied to retain only alphabetic characters and necessary whitespace.

4.1.5 Stripping Whitespace

Any leading or trailing spaces in the tweets were removed. This ensures that the text is properly formatted and avoids inconsistencies during tokenization and further processing.